Rama KrishnaUpthala
Data Engineer
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Summary

Data Engineer with 5+ years of experience designing, building, and optimizing scalable, production-grade data pipelines across AWS, Azure, and GCP. Specialized in ETL/ELT development using Spark, SQL, and Python, with deep experience in Azure Data Factory, Databricks, Azure SQL, and ADLS. Proven at owning end-to-end batch and real-time streaming pipelines, processing large-scale datasets, improving performance and cost efficiency, and delivering reliable, high-quality data solutions for analytics, BI, and data science teams.

Technical Skills

- **Programming**: Python, SQL, Java
- **Big Data & Streaming**: Apache Spark, PySpark, Kafka, Spark Structured Streaming, Delta Lake, Hadoop
- **Cloud Platforms**: AWS (S3, Glue, Redshift, Lambda, EMR, RDS, Kinesis), Azure (Data Factory, Databricks, ADLS, Synapse, Azure SQL)
- **Data Warehousing & Databases**: Redshift, Snowflake, Azure SQL, Post-gre SQL, MySQL, NoSQL
- **Orchestration & DevOps**: Airflow, Jenkins, Azure DevOps, Git, CI/CD
- **Monitoring & Governance**: Cloud Watch, Azure Monitor, Unity Catalog, Data Quality Frameworks
- **BI & Analytics**: Power BI, Looker, Tableau

Professional Experience

KaiserPermanente Health Atlanta, GA **May2024 – Present**
Azure Data Engineer

Technologies: Azure Data Factory, Azure Databricks, Azure SQL, ADLS, Python, Py-Spark, Azure Monitor, Logic Apps, Airflow, Power BI

- Designed and owned **Azure-based batch and streaming data pipelines** using Azure Data Factory and Databricks to support enterprise healthcare analytics and reporting.
- Developed **Py-Spark ETL pipelines** to process large-scale healthcare datasets in ADLS, reducing end-to-end processing time by **25%**.
- Modeled and optimized **Azure SQL data models** through indexing and query tuning, improving query performance by **20%** for downstream analytics workloads.
- Implemented and managed a **centralized Azure Data Lake (ADLS)** to support analytics, reporting, and regulatory compliance requirements.
- Integrated **machine learning and generative AI models** into production data pipelines to enable predictive analytics and advanced insights.
- Automated pipeline orchestration, monitoring, and alerting using **Azure Monitor, Logic Apps, Airflow, and Jenkins**, improving reliability and reducing operational overhead.
- Enabled self-service analytics by delivering **clean, reliable datasets** consumed by Power BI, Looker, and Tableau dashboards.
- Collaborated closely with **data science, analytics, and business stakeholders** to ensure data accuracy, usability, and alignment with business requirements.

Lowe's- Alpharetta, GA **Jan2023 – Apr2024**

AWS data Engineer

Technologies: AWS Glue, S3, Redshift, Lambda, Kinesis, RDS, Spark SQL, Python, SQL, CloudWatch, Kafka

- Designed and owned **scalable AWS-based ETL pipelines** using AWS Glue and Python to ingest, transform, and deliver enterprise retail data for analytics and reporting.
- Architected and optimized **Amazon S3 data lake structures** using partitioning and compression, reducing storage costs by **25%** while improving query efficiency.
- Developed and optimized **Amazon Redshift data warehouse schemas**, improving query performance and accelerating reporting for downstream analytics teams.
- Implemented **serverless data ingestion workflows** using AWS Lambda and Glue, reducing operational overhead and improving pipeline scalability by **20%**.
- Built and maintained **real-time and near real-time streaming pipelines** using AWS Kinesis and Kafka to support time-sensitive analytics use cases.
- Authored **complex SQL and Spark SQL transformations** to aggregate and process large-scale datasets for analytical consumption.
- Implemented **monitoring and alerting** with Amazon CloudWatch to ensure pipeline reliability, fast incident response, and SLA adherence.

D-Mart Hyderabad, India **Mar2021-Jul2022**
Big Data Engineer

Technologies: AWS (S3, RDS, Redshift), Apache Spark, PySpark, SQL, NoSQL, Python, Pandas, Airflow, Scikit-learn

- Built and maintained **scalable big data ETL pipelines** using Apache Spark, Py-Spark, and Airflow to ingest data from transactional systems, logs, and external sources into AWS data lakes.
- Designed and optimized **batch processing workflows** on Amazon S3 and Redshift to efficiently process large datasets and support analytics and reporting use cases.
- Developed Spark-based data transformations to clean, join, and aggregate datasets, reducing overall data processing time by **25%**.
- Designed and optimized **SQL and NoSQL data models** to support high-volume analytical queries and near real-time insights.
- Implemented **data quality checks and validation frameworks** using Python and Pandas, reducing missing, duplicate, and inconsistent records by **35%**.
- Delivered **curated, analytics-ready datasets** for business and marketing teams, enabling campaign analysis and contributing to a **40% improvement in website traffic** through data-driven insights.
- Built and maintained **ML-ready data pipelines** and feature datasets to support churn prediction models using Scikit-learn.
- Tuned performance of **AWS S3, Redshift, and RDS** by optimizing partitions, indexing, and query patterns, reducing storage and compute costs by **20%**.
- Collaborated with cross-functional teams to gather requirements, troubleshoot pipeline issues, and ensure reliable delivery of big data solutions.

Education:

University of Bridgeport **Sep 2022-Apr 2024**

Master of Science,computerscience ***G*PA:**3.59

Certifications:

- **Data Engineering:** With Hadoop and Spark certification and python certification from Geeks for geeks
- **AWS Academy Graduate:** AWS Academy Cloud Foundation certification by AWS